

6.2.4 Electric potential and energy

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) electric potential at a point as the work done in bringing unit positive charge from infinity to the point; electric potential is zero at infinity	
(b) electric potential $V = \frac{Q}{4\pi\epsilon_0 r}$ at a distance r from a point charge; changes in electric potential	
(c) capacitance $C = 4\pi\epsilon_0 R$ for an isolated sphere	Derivation expected from equation for electric potential and $Q = VC$.
(d) force–distance graph for a point or spherical charge; work done is area under graph	HSW5
(e) electric potential energy $= Vq = \frac{Qq}{4\pi\epsilon_0 r}$ a distance r from a point charge Q .	